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Hovering Helicopter Rotors Modeling Using the Actuator Line Method

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An implementation of the actuator line method (ALM) is applied to a hovering helicopter rotor. This method, which is widely used for wind turbine simulations, replaces the rotor blades by momentum source terms in the unsteady Reynolds-averaged Navier–Stokes equations. The removal of the blade mesh significantly reduces the computational mesh size, thus lowering the computational cost. The ALM is presented along with some improvements, notably the choice and treatment of the projection kernel. A parameter sweep is performed showcasing the importance of proper selection of the Gaussian smearing coefficient ϵ for accurate rotor performance predictions with a value of ϵ scaled around a quarter chord in size. With this value, a new set of simulations on a refined mesh is performed and analyzed covering global rotor performance coefficients, sectional blade loading, tip vortex characteristics in terms of positions, circulation, and core radius. The ALM is benchmarked against an equivalent blade resolved case on the well-known S-76 rotor. Results confirm the appropriateness of the ALM model for a hovering rotor for main flow features and performance metrics, although there was a small loss of accuracy on the tip blade loading in the presence of a blade–vortex interaction. Finally, computational performances indicate an elapsed-time speed-up between 3 and 4× in addition to a greater parallel efficiency in favor of the ALM.

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Nomenclature

A_{disk}	=	area of the rotor disk, m^2
AR	=	aspect ratio, R/c_{ref}
C_d	=	blade section 2D drag coefficient
C_l	=	blade section 2D lift coefficient
C_Q	=	rotor torque coefficient, $Q/(\rho_\infty {V_{\text{tip}}}^2 A_{\text{disk}} R)$
C_T	=	rotor thrust coefficient, $T/(\rho_\infty {V_{\text{tip}}}^2 A_{\text{disk}})$
c	=	local chord, m
c_e	=	thrust weighted equivalent chord, $3\int_0^1 c r^2 \mathbb{d}r$, m
c_{ref}	=	reference chord, m
$\boldsymbol{f}_{\text{alm}}$	=	volumetric body force of the actuator line method
g_{iso}	=	isometric Gaussian kernel
g_{tr}	=	truncated Gaussian kernel
M	=	Mach number
N_b	=	number of blades
R	=	rotor radius, m
R_{root}	=	rotor root cut out radius, m
Re	=	Reynolds number
r	=	radial coordinate along span normalized by R
r_c	=	vortex core size, m
r_v	=	vortex core radial coordinate, m
$\boldsymbol{U_P}$	=	perpendicular velocity to the rotor disk, m/s
$\boldsymbol{U}_{\text{rel}}$	=	relative velocity a blade section sees, m/s
$\boldsymbol{U_s}$	=	sampled velocity at kernel center, m/s
$\boldsymbol{U_T}$	=	tangential velocity to the rotor disk, m/s
V_{tan}	=	tangential velocity of vortex core, m/s
$\bar{V}_{\text{tan}}$	=	average tangential velocity across vortex core, m/s
V_{tip}	=	velocity of the blade tip, m/s
z	=	rotor axial axis coordinate normalized by R
α_{rel}	=	relative angle of attack of a blade section, deg
β_{norm}	=	volumetric normalization factor
Γ	=	circulation of a vortex core, m^2/s
ΔD	=	blade segment drag force, N
ΔL	=	blade segment lift force, N
ΔR	=	blade segment radial length, m
ϵ	=	Gaussian smearing width parameter
Θ_{75}	=	collective pitch at 75% blade span, deg
σ	=	rotor solidity, $N_b c_{\text{ref}}/\pi R$

Subscript

i	=	actuator line method node index on the rotor blades

I. Introduction

HELICOPTERS are used to access hard-to-reach destinations and operate in the close vicinity of complex terrain and obstacles. In addition, the geometry of these obstacles can be the source of complex 3D flow conditions, which pose an added challenge to the workload of the pilot. One such example is the landing of a helicopter on the deck of a ship at sea, which is particularly difficult for the pilots especially in high-crosswind conditions [1]. To help in the training of pilots in such situations, helicopter aircraft simulators must be able to recreate these challenging conditions, which necessitate a high level of realism and comprehensive physical models. High-fidelity numerical simulations are typically used to augment the physical modeling of the aircraft simulator through calibration or by using wind maps extracted from the flowfield in some coupling procedures [2–5].

Different methods exist for simulating the aerodynamic performances of helicopter rotors. These techniques vary in terms of level of fidelity along with their respective computational cost having orders of magnitude difference between them. Simple models such as blade element theory (BET) and blade element momentum theory (BEMT) [6] are extremely fast in terms of computational time and are often at the core of real time flight simulator systems, but often lack accuracy in challenging flow conditions such as in ground effect or in interaction with walls where flow recirculation can exist and affect the performance of the rotor [7]. Extensive and specific modeling would be needed to increase the accuracy of the models, and they would still need to be fine-tuned and benchmarked against higher-fidelity models or experimental data. Medium-fidelity models, most notably singularity-based vortex methods [8–10], offer an increased level of fidelity at a still reasonable compute time. By modeling a subset of the fluid flow equations, a reasonable level of accuracy can be obtained in hovering conditions or in interaction with simple walls or floor geometries. These methods become less than ideal as the interacting geometries become more complex and are often plagued

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by numerical instabilities caused by their singular nature. Finally, high-fidelity meshed-based Navier–Stokes/Euler simulations [11–13] that model the exact geometry of the rotor and the surrounding obstacles using body-fitted grids yield more accurate results as the equations are more complete and the geometry is explicitly included in the numerical models. Their main drawback is the setup and computational time that can take weeks to compute on large computer clusters, which are orders of magnitude more when compared to other lower-fidelity techniques mentioned earlier.

This paper focuses on the actuator line method (ALM) introduced by Sørensen and Shen [14] in the wind energy sector as an evolution of the steady actuator disk (AD) technique, which has been present in rotor flow simulations for the past 30 years [15–17]. The ALM, along other "actuator-type" methods, is an alternative to traditional fully blade resolved (B-R) unsteady Reynolds-averaged Navier–Stokes (U-RANS) simulations that eliminates the blades geometries and thus their associated body-fitted meshes. This significantly reduces the mesh size of the flow domain and eliminates the possible Chimera/overset mesh overhead, which further reduces computational time. The effect of the rotor blades is replaced by momentum source terms that exert forces on the flowfield resulting in similar flow characteristics. The integrated performance coefficients are recovered along with the blade loading and the full wake details of the discrete set of tip vortices.

The ALM is seldom used in the helicopter community. Forsythe et al. [18] introduced a slightly modified version from the original model by sweeping the force source term to achieve larger time steps toward applying it to a helicopter landing on a ship. Schmitz and Jha [19] used it as a pure drag device to model an isolated rotor hub. Stanly et al. [20] recently showed preliminary results using the ALM with the aim of computing rotor noise. Alpman et al. [21] coupled a helicopter flight dynamic model to a modified ALM model for the computation of rotor load time histories for a rotor placed behind a hangar with incoming wind as well as a rotor behind a ship frigate to evaluate different coupling procedures. Other similar rotor replacement techniques exist and are being used and researched in the helicopter community in recent years. The unsteady actuator disk (UAD) was used to study rotor–fuselage interaction by Boyd [22], and the interaction between a rotor and a ship frigate in a simulation coupled with a flight dynamic model by Oruc et al. [5], whereas Chirico et al. [23] used it to investigate the interaction between a rotor and a generic box-shaped obstacle. An actuating blade introduced by O'Brien and Smith [24] was used to study rotor–fuselage interaction and was benchmarked against AD and Chimera B-R solutions. The actuator surface method (ASM) was used by Kim and Park [25,26] to study the interaction between a rotor and a simple fuselage geometry as well as a test case in hover. Wood et al. [27] used a simplified ASM model with a constant loading to study wake breakdown in high-resolution hovering simulations on the S-76 geometry. Kim et al. [28] proposed an improved ASM that couples the model to vortex panels to obtain a more accurate induced velocity at the blade and validated their results on a rotor in a hovering condition and in forward flight with a fuselage. Linton et al. further developed the method [29] and later incorporated an unsteady aerodynamic model [30] to take into account the time history in the force response for the case of a parallel blade–vortex interaction (BVI) and of an oscillating airfoil.

For helicopters, the motivation behind these new rotor replacement techniques is to have a more computationally efficient approach than the traditional B-R simulations, making them an ideal tool for complex rotor interaction studies such as with fuselages or obstacles. This efficiency comes at the cost of some accuracy, which can be an acceptable tradeoff when only main flow features and their interactions are of interest and not the fine details of the flowfield and blade surface coefficients. The majority of the rotor replacement techniques discussed above are often used directly on complex interactional cases and sometimes validated succinctly on hovering cases. Furthermore, in recent years, a substantial effort was made in the helicopter community to validate the predictive performances of various computational fluid dynamics (CFD) codes for a rotor in hovering conditions through the AIAA Hover Prediction Workshop (HPW) [11].

To better understand the current predictive capabilities of this class of methods, this work explores an in depth validation of the ALM in hovering conditions. This method is chosen in favor of the others because of its simplicity attributed to its line-like representation of the force and to less stringent mesh requirements compared to an ASM. Furthermore, it is the most popular method used in the wind energy sector for horizontal axis wind turbines, which gives an additional wealth of information and advancements through recent research in that sector. The method is presented along with a set of modifications performed with respect to the original method for more consistent, robust, and accurate results. The analysis and performance metrics are also in tune with what was done during the HPW. The first part of the results explores the effect of the proper selection of the force smearing coefficient ϵ and its effects on the integrated performance plots, blade loading, and tip vortex positions for a full collective sweep. Then the second part of the results compares an ALM solution to a B-R solution on these same metrics and performs a more detailed analysis of the tip vortex properties at an equal trim level. Finally, the computational efficiency of the two methods is compared.

II. Test Case and Rotor Geometry

The selected geometry for this paper is the 1/4.71 model scale of the S-76 rotor used in the experimental study of Balch and Lombardi [31,32]. This geometry has been used as the central test case of the HPW due to its public availability and now offers a wide variety of numerical comparative studies. The characteristics of the rotor are listed in Table 1. The geometry of the different sections along the span is defined at specific breakpoints presented in the table. Between the breakpoints the geometry is linearly interpolated for the B-R simulations, whereas the blade section properties are linearly interpolated for the ALM. The airfoil geometries are taken from Bousman [33]. The SC1094-R8 airfoil is manually scaled to 13 and 10.09% thickness to obtain the airfoil profiles of the first two radial sections, respectively. The last 5% of the span toward the tip is where the different tip shapes usually differ, but this work focuses only on the rectangular tip variant of the S-76 since the other tip shapes all exhibit similar blade loading characteristics with only subtle changes present near the tip region as shown in Ref. [34]. Furthermore, swept blade tip geometries create a crossflow component that needs to be taken into account by a modified 2.5D airfoil database and coupling algorithm as per Refs. [35,36] that are not yet incorporated in this ALM implementation.

On this geometry, one tip Mach number of $M_{\text{tip}} = 0.60$ is considered in this work with an associated Reynolds number of $Re_{\text{tip}} = 1.1 \times 10^6$. This corresponds to the Test Run 54 in the experimental reports of Balch and Lombardi. The ALM simulations are run at collective angles $\Theta_{75} = 4.5°$ through 9.5° by increments of 1°, which gives the most direct comparison points with the experimental data.

Table 1 S-76 model rotor characteristics

Rotor definition		Blade sections definition			
Parameter	Value	Section (r/R)	Twist (°)	Thickness (t/c%)	Airfoil
No. of blades (N_b)	4	0.189	4.01	13	SC1013-R8
Rotor radius (R)	56.04 in.	0.285	4.5	10.09	Blend
Reference chord (c_{ref})	3.1 in.	0.400	3.5	9.4	SC1094-R8
Aspect ratio (AR)	18.077	0.750	0	9.4	SC1094-R8
Solidity (σ)	0.07043	0.800	−0.5	9.4	SC1094-R8
Linear twist	−10°	0.840	−0.9	9.5	SC1095
		0.950	−2	9.5	SC1095
		1.000	−2.5	9.5	SC1095

The B-R simulations are limited to collectives of $\Theta_{75} = \{6.5°, 7.5°, 9.5°\}$ due to their higher computational costs. An additional set of B-R and ALM simulations is run on a finer mesh at a trimmed state of $C_T/\sigma = 0.09$ for a more accurate comparison of blade loading and wake characteristics.

III. Numerical Modeling

A. Flow Solver

The software used in this work is Star-CCM+ 12.06 [37], which is a general-purpose finite-volume, unstructured, cell-centered, multiphysics flow solver. All computations are done with a coupled flow solver (density based, compressible) with an unsteady dual-time-stepping approach with a second-order accuracy in time. Spatial discretization uses a second-order upwind scheme with inviscid fluxes evaluated by a Weiss–Smith preconditioned Roe flux-difference-splitting scheme [38]. The U-RANS equations are solved with the $k-\omega$ SST turbulence model [39] as a closure. The ALM is implemented through the User Coding framework and loaded as a shared library into Star-CCM+, which applies the resulting field function as a momentum body force source term in the flow equations. The library is written in C and parallelized with the same MPI implementation as the base flow solver for communications. For faster convergence and reduced domain size, a source-sink-type boundary condition is adapted from Mohd and Barakos [40] and Vieira et al. [41] for the velocity inlet and outlet boundary conditions used in Star-CCM+.

B. Blade Resolved Simulations

The B-R simulations model the four rotating blades as four different overset mesh regions interfaced with a static background mesh. No rotor hub or linkages are considered or modeled. The interpolation between the different overset mesh levels is done with an inverse distance weighted approach. These rotating mesh regions are cylinder shaped and encompass their respective blades, which are then put into motion by applying a rotational velocity on the geometry that then moves the blade meshes at every completion of the flow solver time step. The chosen time step corresponds to 1° of azimuthal rotation. The only exception is the B-R simulation at a trimmed state of $C_T/\sigma = 0.09$, which required the time step to be reduced to 0.25° due to its refined mesh. The dual-time-stepping approach uses 15–20 subiterations per time steps for the flow residuals to drop approximately three orders of magnitude or more at the end of each physical time step.

C. ALM Modeling

The ALM is a model that allows the computation of the flow around rotors by removing the blade geometries from the computations, therefore only leaving the background mesh for resolving the wake. Thus, a substantial amount of grid points can be eliminated in addition to the removal of overset mesh overhead resulting in faster computation and setup time. The rotor forces are computed by the model and applied to the flowfield through momentum source terms in the fluid flow equations. These equations are summarized in their generic integral form, omitting any turbulence model, by Eqs. (1–3), as per Ref. [37]:

$$\frac{\partial}{\partial t}\int_V \rho \mathrm{d}V + \oint_A \rho \boldsymbol{v}\cdot \mathrm{d}\boldsymbol{a} = 0 \tag{1}$$

$$\frac{\partial}{\partial t}\int_V \rho \boldsymbol{v}\mathrm{d}V + \oint_A \rho \boldsymbol{v}\otimes\boldsymbol{v}\cdot \mathrm{d}\boldsymbol{a} + \oint_A p\boldsymbol{I}\cdot \mathrm{d}\boldsymbol{a} = \oint_A \boldsymbol{T}\cdot \mathrm{d}\boldsymbol{a} + \int_V \boldsymbol{f_b}\mathrm{d}V - \int_V \boldsymbol{f_{alm}}\mathrm{d}V \tag{2}$$

$$\frac{\partial}{\partial t}\int_V \rho E\mathrm{d}V + \oint_A \rho H\boldsymbol{v}\cdot \mathrm{d}\boldsymbol{a} + \oint_A \boldsymbol{q}\cdot \mathrm{d}\boldsymbol{a} = \oint_A \boldsymbol{T}\cdot\boldsymbol{v}\mathrm{d}\boldsymbol{a} + \int_V \boldsymbol{f_b}\cdot\boldsymbol{v}\mathrm{d}V - \int_V \boldsymbol{f_{alm}}\cdot\boldsymbol{v}\mathrm{d}V \tag{3}$$

where V is the control volume of a cell and $\boldsymbol{a}$ its outward area vector of a given face. Flow properties are represented by ρ the density, $\boldsymbol{v}$ the velocity vector, p the pressure, $\boldsymbol{T}$ the viscous stress tensor, $\boldsymbol{q}$ the heat flux vector, and E and H the total energy and enthalpy, respectively. The term $\boldsymbol{f_b}$ represents a generic body force such as gravity or centrifugal force and is taken into account in the energy equation through the scalar product with the velocity vector. The volumetric body force needed for the modeling of the ALM is represented by the $\boldsymbol{f}_{\mathrm{alm}}$ term, which is treated as any other body force. The negative signs indicate that the force to be applied on the flowfield is of equal strength, but opposite direction of the rotor forces. In this work, the energy source term in Eq. (3) is omitted as it has only a minor impact on the results and saves computational time (results not shown for the sake of brevity). The energy equation is more often than not neglected in wind turbine simulation as a result of the incompressible flow equations that are solved. In [42], it was also shown that this term had no significant impact on the results for subcritical cases.

To model the $\boldsymbol{f}_{\mathrm{alm}}$ source term, a BET approach is taken and the process is illustrated in Fig. 1. The different blades are treated as lines

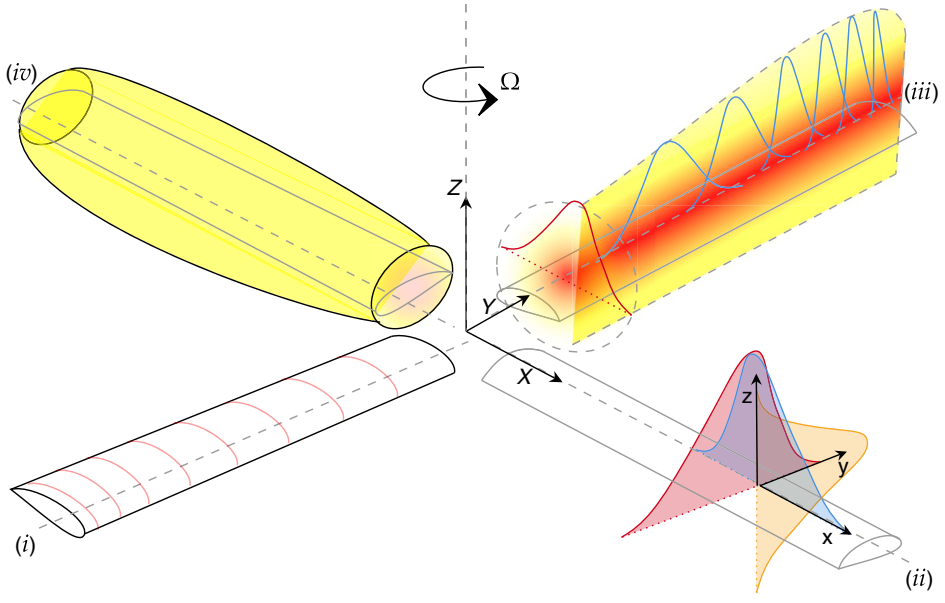


Fig. 1 ALM representation with different blades representing different steps in the process. i) Blade element section discretization. ii) Gaussian smoothing in three spatial dimensions. iii) Chordwise and spanwise cut planes of the body force. iv) Body force isosurface.

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and discretized by points along their span axis corresponding to the local quarter chord point of the profile. The points form line segments with the flow properties evaluated at the center of each line segments and form an ALM control point. The velocity is sampled at this point, denoted by $\boldsymbol{U}_s$, and represents the wake-induced velocity void of any bound circulation-induced velocity. More details on the velocity sampling will follow. With the velocity known, the local equivalent freestream angle of attack α_{rel} and velocity $\boldsymbol{U}_{\mathbf{rel}}$ are reconstructed as per Eqs. (4–7) and illustrated in Fig. 2. The identification of these two properties allows the interpolation in a 2D viscous database generated in Star-CCM+.

$$\boldsymbol{U_T} = \Omega(r \cdot R) - \boldsymbol{U}_{s,y} \tag{4}$$

$$\boldsymbol{U_P} = -\boldsymbol{U}_{s,z} \tag{5}$$

$$\boldsymbol{U}_{\mathbf{rel}} = \boldsymbol{U_P} + \boldsymbol{U_T} \tag{6}$$

$$\phi = \arctan(U_P/U_T) \tag{7}$$

The database in this work comprises 10 sections along the span where lift, drag, and moment polars are evaluated at angles of attack ranging from −10° to 17°. This wide angle-of-attack range ensures that all the simulations in this work have their effective angle of attack well within the database range. Otherwise, the values of the coefficients at the end of the range would be used and extrapolated. The geometry and flow properties, such as Mach and Reynolds numbers, change according to their respective radial positions along the span. With the information about the local effective velocity and the known force coefficients, the force of every sections can be redimensionalized as per Eqs. (8) and (9) and projected onto the flowfield.

$$\Delta L = \frac{1}{2}\rho\|\boldsymbol{U}_{\text{rel}}\|^2 C_l(\alpha_{\text{rel}}, Re, M)c\Delta R \tag{8}$$

$$\Delta D = \frac{1}{2}\rho\|\boldsymbol{U}_{\text{rel}}\|^2 C_d(\alpha_{\text{rel}}, Re, M)c\Delta R \tag{9}$$

The force is then spread over several grid cells through a projection kernel. The one used in this work is a modified version of the isometric Gaussian kernel typically used in ALM simulations shown by Eq. (10). Every ALM control point is centered on the section quarter chord point (x_0, y_0, z_0) and has its own kernel associated to it as shown in Fig. 1 on blade (ii). Then, $\boldsymbol{f}_{\text{alm}}$ is constructed by summing all the products of the kernels multiplied by their associated force vectors as illustrated on blade (iii) and shown in Eq. (13), where $\boldsymbol{e_L}$ and $\boldsymbol{e_D}$ are the unit vectors representing the direction of application of the sectional lift and drag force, respectively. These forces are later converted to a global coordinate system taking into account the induced flow and angle of attack.

$$g_{\text{iso}}(x, y, z) = \frac{1}{\epsilon^3\pi^{\frac{3}{2}}}\exp\left(-\frac{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2}{\epsilon^2}\right) \tag{10}$$

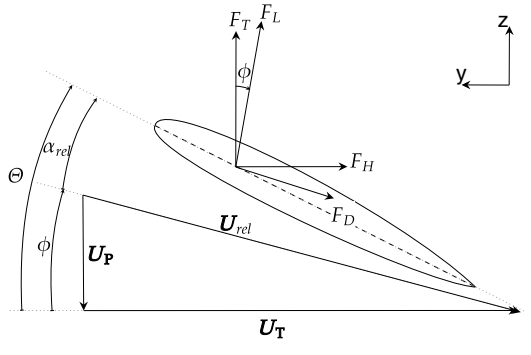


Fig. 2 Blade element representation of a blade section.

$$g_{\text{tr}} = \begin{cases} g_{\text{iso}}, & \text{if } R_{\text{root}} \le \sqrt{(X)^2 + (Y)^2} \le R_{\text{tip}} \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

$$\beta_{\text{norm}} = \frac{1}{\int_V g_{\text{tr}}\,\mathrm{d}V} \tag{12}$$

$$\boldsymbol{f}_{\text{alm}} = \sum_i \beta_{\text{norm},i} \cdot g_{\text{tr},i} \cdot (\Delta L_i \boldsymbol{e_{L,i}} + \Delta D_i \boldsymbol{e_{D,i}}) \tag{13}$$

Some improvements to the traditional ALM are performed and detailed as follows. The kernel is truncated at both the tip and the root and normalized as shown in Eqs. (11) and (12) and illustrated in Fig. 1 on blade (iv). Truncating the kernel constrains the source term to respect the geometrical limits of the blade in the spanwise direction as no force is projected past the root or tip of the blade. If the Gaussian projection is not truncated, the source terms would project force past the root and tip of the blade, whereas varying the smearing parameter ϵ would modify the distance at which this overprojection would end. Both the overprojection past the root and tip and its variable distance would modify the force distribution, which results in an inconsistent tip vortex position and strength. The normalization allows to conserve the property of the kernel to integrate to unity, which is not the case when truncating it or when using a coarser mesh. It is to be noted that although the truncation and normalization are applied to every kernel for every ALM control points, only the ones close to the root and tip are considerably affected because a typical simulation has its smearing parameter scaled at $\epsilon \sim 25\%$ of the local chord c and the kernel decays quickly to ∼0 within 3ϵ. For reference, integrating the Gaussian spread force over a 3ϵ span corresponds to 99.9934% of the total projected force.

For a better force representation along the span of the blade, the ALM control points follow an hyperbolic tangent distribution. This distribution allows a much finer ALM discretization near the tip, which sees the greatest variation of blade loading, while keeping the overall number of ALM points low. Other distributions were tried, most notably the cosine distribution, but it was deemed too aggressive in its refinement ratio. The hyperbolic tangent allows a more gradual and controllable refinement as per Eq. (14), where χ is the normalized [0, 1] coordinate, x is an array of equally spaced coordinate that spans the range $[0, \zeta]$, and ζ is the tuning parameter:

$$\chi = \frac{\tanh(x)}{\tanh(\zeta)} \tag{14}$$

In the present work, the rotor is modeled with 65 ALM control points per blade and $\zeta = 1.375$. This discretization ensures the convergence of the ALM model for the present geometry and test cases.

Different approaches exist to evaluate the velocities at the ALM control points. Merabet and Laurendeau [42] identified the integral velocity sampling proposed by Forsythe et al. [18] and Churchfield et al. [43] as being the most robust and overall most accurate method in a 2D framework. Caprace et al. [44] further generalized the sampling technique to a 3D lifting line. The technique relies on the same kernel projection function used to project the force and essentially acts as a weighted average of the velocity field multiplied by the kernel times the volumes of the cells as shown in Eq. (15).The analytical developments of the integral velocity sampling ensure that any kernel that integrates to unity can be used; hence it is surmised that this method, developed in 2D, is well suited for the present truncated-normalized Gaussian kernel for most of the span with possible exceptions near the tip and root where 3D effects are stronger due to the tip and root vortices. The method thus samples the wake-induced velocity $\boldsymbol{U}_{s,i}$ directly at the ALM control point located at the quarter chord point. No correction for bound circulation is needed and all the information about the wake induced velocity is present. It then becomes straightforward to reconstruct a velocity triangle and recover $\boldsymbol{U}_{\text{rel}}$ and α_{rel} from the measured induced velocity, blade rotational speed, and blade geometry as per BET.

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$$\boldsymbol{U}_{s,i} = \int_V \beta_{\text{norm},i} \cdot g_{\text{tr},i} \cdot \boldsymbol{v} \mathrm{d}V \tag{15}$$

In most ALM implementations, the model uses a tip-loss function as it was noticed that the blade loading near the tip would grow to abnormally high values, thus resembling an uncorrected BEMT solution. But this correction should not be needed in a 3D U-RANS framework as the tip vortices are resolved as noted in [43]. Given proper shed vorticity and tip vortex structures, the ALM should not need to be corrected artificially because the induction of the wake is correctly captured by the method. The value of the smearing parameter ϵ greatly affects the tip vortices, which in turn affects the blade loading and performance coefficients. The general tendency in wind turbine simulations is to use higher values of ϵ as it allows a coarser mesh resolution. Several researchers [45–47] identified an optimal value of the smearing coefficient to be in the area of $\epsilon/c = 0.25$ as this value produced the best results and does not need a tip-loss function to produce accurate blade loading distributions. An in-depth look at this issue can be found in Martinez-Tossas and Meneveu [47]. Other Gaussian smearing coefficient guidelines exist [48,49], but they pertain to more coarsely resolved wind turbine simulations. Therefore, this paper considers the aforementioned value of the Gaussian smearing parameter as the de facto value along with no tip-loss function. Different values of ϵ are still used on a coarse mesh to assess the effects of the parameter.

Finally, more implementation specific improvements are used in this work. For an improved conditioning of the startup revolutions and faster thrust convergence, the force is ramped up from zero to its nominal value over the first two revolutions. An under-relaxation of the applied force is also implemented for the intra time-step subiterations. This under-relaxation ensures an increased convergence of the method that tends to be noisy on coarse meshes or as small values of ϵ are used. An under-relaxation factor of 0.75 typically produces a normalized convergence of the thrust and torque coefficients of at least four orders of magnitude on the last subiteration of the time step.

Table 2 Mesh cell sizes

	Refinement zone			Blade mesh	
Grid	Near wake	ALM near field	T.V.	No. of prism layers	Surface size
B-R coarse	25.0% c_{ref}	— —	— —	20	4.0% c_{ref}
B-R medium	17.5% c_{ref}	— —	— —	30	4.0% c_{ref}
B-R T.V.	17.5% c_{ref}	— —	4.0% c_{ref}	30	4.0% c_{ref}
B-R fine	12.5% c_{ref}	— —	— —	30	2.5% c_{ref}
ALM coarse I	25.0% c_{ref}	12.5% c_{ref}	— —	— —	— —
ALM coarse II	30.0% c_{ref}	7.5% c_{ref}	— —	— —	— —
ALM medium	16.0% c_{ref}	8.0% c_{ref}	— —	— —	— —
ALM T.V.	16.0% c_{ref}	8.0% c_{ref}	4.0% c_{ref}	— —	— —
ALM fine	11.7% c_{ref}	5.8% c_{ref}	— —	— —	— —

D. Computational Meshes and Refinement

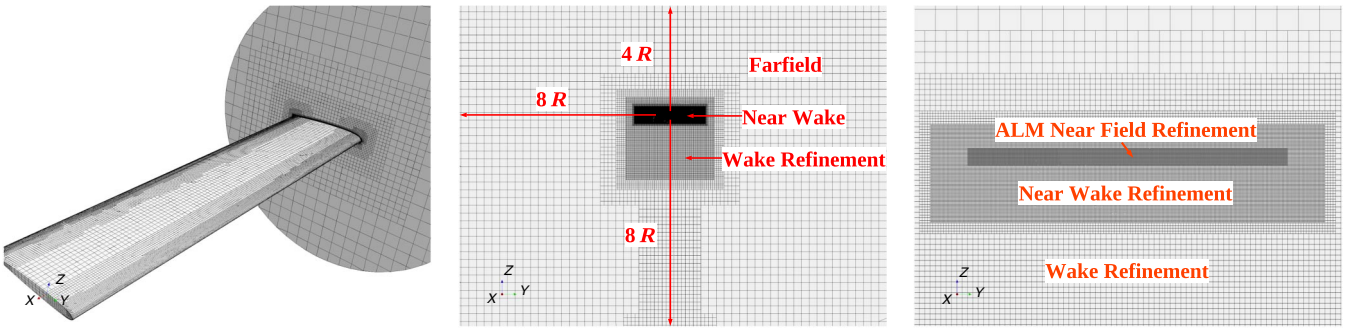
For a fair comparison between the B-R and the ALM, the two methods share similar background meshes. The difference lies with the B-R needing every blade to be meshed individually and the ALM needing a finer mesh around the rotor disk for proper resolution of the source terms especially at lower ϵ values. Flow domains are meshed with cubic hexahedron cells in a oct-tree-type mesh.

Every simulation is run in a cylindrical background mesh that extends $4R$ above the rotor origin, whereas the outflow and radial boundaries are $8R$ from the origin. Successive cylindrical refinement zones are placed near the rotor disk region to better capture the near wake. The finest refinement zone of the mesh, i.e., "near wake" is a cylinder slightly larger than the rotor radius R and spans 0.25 and 0.6 m above and below the rotor disk, respectively. A background mesh for an ALM simulation is displayed in Fig. 3b. These dimensions ensure that at least one revolution of the wake is captured at the highest collective setting. The ALM has a further refinement zone, i.e., "ALM near field" that spans one chord above and below the rotor disk as shown in Fig. 3c. The sizes of the cells in all the meshes used are reported in Table 2.

The B-R simulations have their four blades individually meshed and interfaced with the background mesh through the overset module. The blade mesh is a cylinder around the blade geometry with a radius of 0.15 m and extends 0.1 m further than the tip and root as displayed in Fig. 3a. The cell size at the boundary of the cylinder matches the background grid cell size. The blade surfaces have their default mesh size listed in Table 2 and local refinement near the tips, leading and trailing edges refined down to 0.5–1% c_{ref}. The boundary-layer height is fixed at 4% c_{ref}, and the amount of cells contained within it is also present in the table.

For the simulations at the trimmed state of $C_T = 0.09$, the tip vortices and their characteristics are looked in more depth and a finer mesh is used in their vicinity. The tip vortex "T.V." refinement zone consists of an annulus. The inner and outer diameters capture the radial position of the tip vortex while its depth is enough to capture these tip vortices for 300° of wake age. This additional mesh refinement results in total mesh size of 54.0 and 73.45 million cells for the ALM and B-R meshes, respectively, which is significantly higher than their respective medium mesh template upon which they are based on.

Table 3 presents the results of a mesh refinement study on the three key integrated performance coefficients on the $\Theta_{75} = 7.5°$ case along with their relative error compared to the experimental value. The errors decrease as the B-R meshes are refined. Both the medium and fine grids have a figure of merit (FoM) error of less than 0.5%, whereas the fine mesh results also have excellent agreement for both the thrust and torque coefficients. The ALM, on the other hand, has a more constant error for all integrated performance coefficients with only a small improvement between the coarse and fine grids. Despite the slightly larger error, the method shows robustness in regard to the mesh density as the goal is to ultimately use the least amount of cells as possible. The improved consistency for the ALM meshes mainly comes from the principal limiting factor, in terms of mesh refinement, being the blade mesh for B-R solutions. Coarsening the blade meshes



a) Blade surface mesh b) Far field c) Near field

Fig. 3 Computational mesh template.

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Table 3 Mesh convergence study at $\Theta_{75} = 7.5°$

	Cell count (millions)			Coefficients					
Grid	Back.	Blade	Total	C_T/σ	Δ [%]	C_Q/σ	Δ [%]	FoM	Δ[%]
B-R coarse	4.0	3.5	17.9	0.07255	0.35	0.00573	2.85	0.63991	−2.26
B-R medium	9.1	4.9	28.7	0.07311	1.12	0.00569	2.18	0.65162	−0.47
B-R fine	24.2	8.1	56.6	0.07213	−0.24	0.00558	0.14	0.65147	−0.50
ALM coarse I	3.9	— —	3.9	0.07020	−2.90.	0.00578	3.81	0.60343	−7.83
ALM medium	10.1	— —	10.1	0.07042	−2.60	0.00576	3.41	0.60857	−7.05
ALM fine	25.6	— —	25.6	0.07048	−2.52	0.00577	3.55	0.60856	−7.05
Exp.	— —	— —	— —	0.07230	— —	0.00557	— —	0.65472	— —

significantly degrades the solution, whereas the ALM is less affected by wake-mesh coarsening as the effects of the blade are modeled internally by the method and all simulations use the same high-quality 2D airfoil polars. The error range for the ALM thrust coefficient is within 2.90% accuracy, whereas the torque coefficient is at most 3.81%, and these are considered reasonable for the type of method employed. The FoM error range is higher with an error within 7.83% because the thrust is underpredicted and the torque is overpredicted.

IV. Tip Vortex Analysis

As the near wake of helicopter rotors is of great importance for its performance predictions, the helical tip vortex wake is analyzed. The process of postprocessing data from tip vortices is typically done manually, which can be both time-consuming and error-prone as noted by Jain [34,50]. The tip vortex analysis in this work is automated for both the extraction of the data from the flowfield and the subsequent postprocessing. The present methodology is based on the works of Goerttler et al. [51] and Linton et al. [30]. The extraction of tip vortices' positions and properties is done by extracting constrained 2D planes of the flow properties around the tip vortices. Because the position is not known beforehand, it is first approximated by the wake model prescribed by Kocurek and Tangler [52]. With a thrust coefficient and a desired wake age, a plane is placed in the wake with its center on the guessed vortex position. Its orientation is determined by the wake age where the plane is rotated at this angle relative to the blade from which the tip vortex originates as illustrated in Fig. 4. Through interpolation from the background mesh, flow properties of interest are interpolated on an $N \times N$ evenly spaced grid and exported as a *.csv* file.

The solution is reconstructed in a Python script, which changes the referential frame to one local to the cut plane. Its axes are aligned with the radial direction given by the rotor disk radius, the rotor disk vertical axis oriented in the thrust direction, and the direction opposite to the rotation of the blade. This referential represents the radial-azimuthal-vertical ordinates (r, Ψ, z), which makes it easy to extract the position of the tip vortices as every cut plane lies in a 2D (r, z) plane. From there, the center of the tip vortex is identified through the maximum Q-criterion. With the center identified, radial lines are traced around the vortex core and the velocities are sampled along these lines. The velocities are further transformed to a vortex tangential-radial-axial reference frame for the identification of the core properties. These lines, typically 90 in this work, are then averaged to obtain an averaged tangential velocity profile $\bar{V}_{\tan}$ across a particular vortex core from which the core size is extracted at the location of the highest ($\bar{V}_{\tan}$) and the circulation is given by Eq. (16). As the value of the circulation Γ varies through the viscous vortex core, a suitable location needs to be identified for its evaluation. Because the core resembles a Vatistas [53] vortex core of power $n = 2$ as per Eq. (17), the circulation of the core is evaluated at $2.5r_c$. As shown in Fig. 5, the empirical core model fits the extracted core model well from the B-R simulations at $\theta_{75} = 7.5°$ on the medium mesh that is displayed. The circulation along the viscous core shown in Fig. 5b varies substantially along the core radius. The Vatistas model tends asymptotically to an equivalent inviscid vortex circulation Γ_∞, whereas the B-R extracted data behave similarly until $2.5r_c$. Past this point, the interaction between the tip vortex, the vortex sheet, and the region outside the contracting wake make the sampled circulation increase further and deviate from the theoretical model. By sampling the circulation at this location, it avoids this problem and recovers 98.7% of the Γ_∞ from the Vatistas model. This method allows a more consistent evaluation of the circulation especially as the wake age gets older where the vortex core diffuses. Sampling at a fixed radius could yield an incorrect circulation by either being too small and measuring the circulation inside the viscous core, or too far and have the effects of the helical vortex wake present.

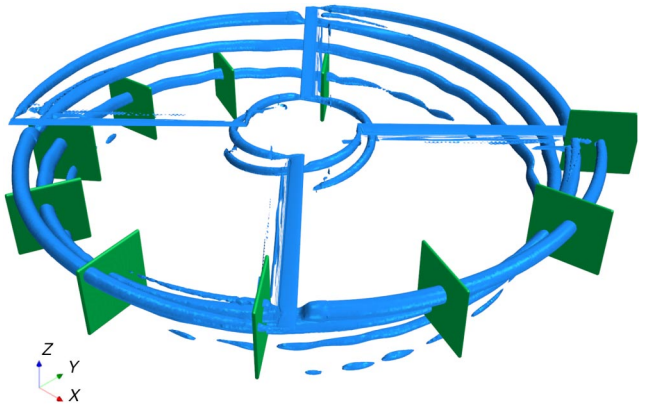


Fig. 4 Tip vortex extraction planes with Q-criterion contour representing the vortices.

$$\Gamma = \oint v \mathrm{d}s = \sum V_{\tan} r \mathrm{d}\theta = 2\pi \bar{V}_{\tan} r \tag{16}$$

$$V_{\tan} = \frac{\Gamma_\infty}{2\pi r_c} \frac{r/r_c}{(1 + (r/r_c)^{2n})^{1/n}} \tag{17}$$

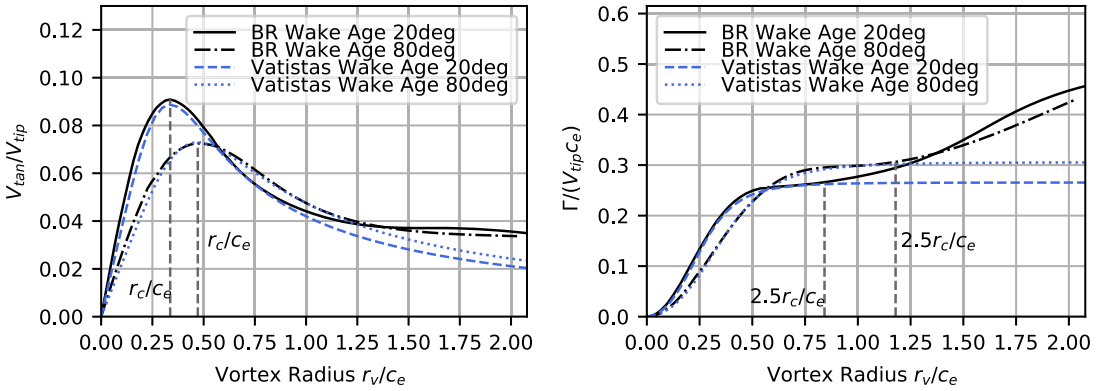
For most of the cases in this work, the meshes used are too coarse to properly resolve the tip vortices, which further adds the drawback of quickly diffusing the viscous core of the vortices. Nonetheless, the position can still be accurately extracted for wake ages under one revolution. Because the background meshes of both the ALM and B-R simulations closely match, the results can be compared together, but are not representative of the real evolution of a tip vortex that would be measured experimentally or computed on a very fine grid as done in [51]. Hence, only the near wake is considered and the comparisons are to be made in between the two methods on similar meshes only. The simulations at the trimmed condition of $C_T/\sigma = 0.09$ in Sec. V.B have a finer mesh for a more in-depth quantitative analysis of the tip vortex properties.

V. Results and Discussion

A. Gaussian Smearing Width Sensitivity

The effects of using different Gaussian smearing parameter ϵ are evaluated in this section with values in the range $\epsilon/c = [0.2, 1.0]$.

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a) Tangential velocity profile **b) Circulation along vortex core**

Fig. 5 Vortex core comparison of B-R extracted data and empirical model.

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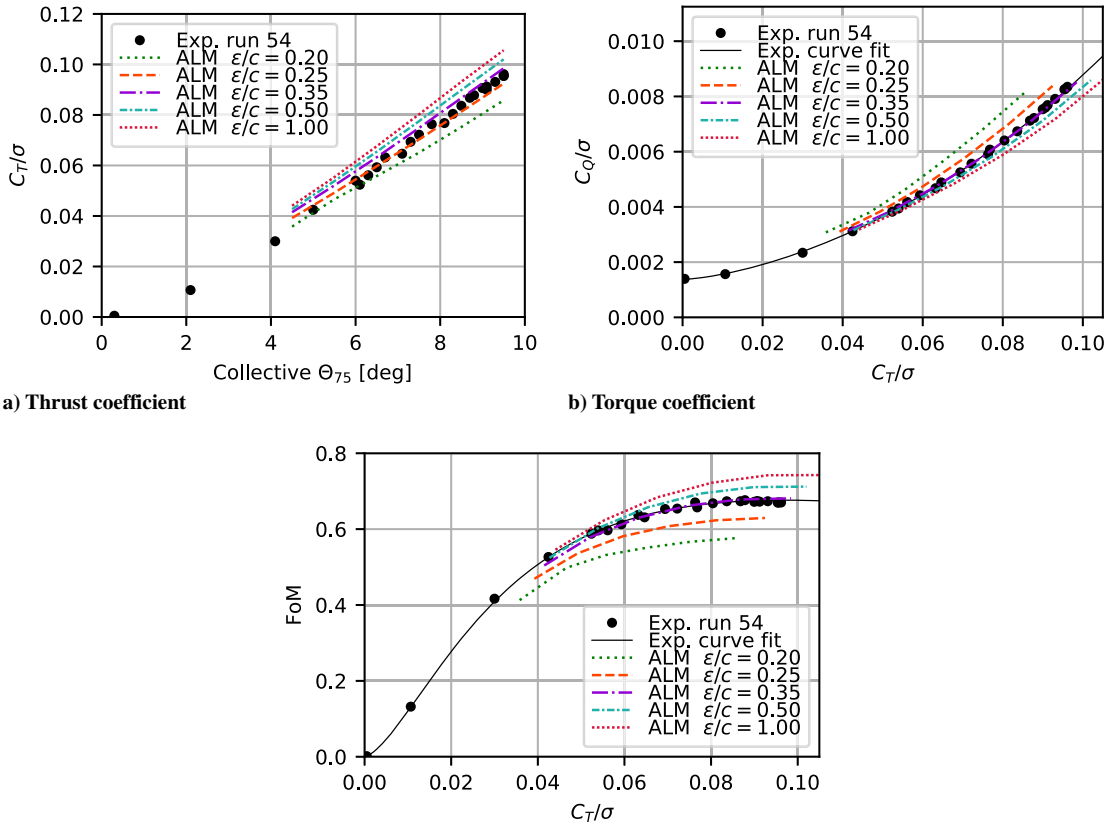
Because of the large amount of simulations, the coarse II mesh is used for the ALM simulations. The results are compared on the basis of their integrated coefficients, blade loading, and tip vortex characteristics.

1. *Integrated Performance Coefficients*

The integrated performance coefficients for the thrust, torque, and FoM are presented in Fig. 6. For a given ϵ, the curves show consistent trends with the experimental data although with some offset. A smaller value of ϵ produces an underprediction of thrust and overprediction of the torque for given thrust level especially at higher collectives. These two errors combined yield an even larger offset for the FoM that is underpredicted. This trend is reversed at the highest values of ϵ as the thrust is overpredicted, whereas the torque is underpredicted at higher collective values, yielding an overprediction of the FoM. From the present figures, $\epsilon/c = 0.35$ produces the best integrated coefficients compared to the experimental data across the whole range of collectives. Then $\epsilon/c = 0.25$ is more accurate for the thrust coefficient across the simulated collective range, especially at the lower range to midrange values, but slightly overpredicts the torque resulting in a lower FoM.

2. *Blade Loading*

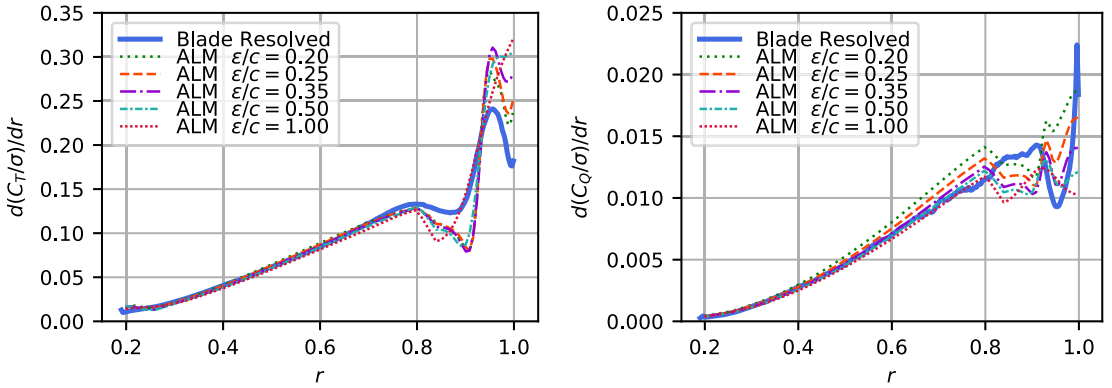
To explain the integrated results, the blade sectional thrust and torque loading are presented in Fig. 7. The ALM simulations are presented at a trimmed thrust of $C_T/\sigma = 0.0723$, which corresponds to the experimental value at $\Theta_{75} = 7.5°$, whereas the B-R is presented at its nominal collective angle of $\Theta_{75} = 7.5°$. The ALM simulations are compared to B-R simulation on the medium mesh because the experimental data did not provide blade loading data. The thrust loading is well approximated from the root until 80% of the span of the blade for all the ϵ values. Although a small constant offset is observed across the range of kernel width with a trend that sees the lower values of ϵ have a marginally higher thrust loading in the first



a) Thrust coefficient **b) Torque coefficient**

c) Figure of merit

Fig. 6 Integrated performance coefficients for the ALM on the coarse II mesh.



a) Thrust loading **b) Torque loading**

Fig. 7 Sectional blade loading of the ALM at $C_T/\sigma = 0.0723$ for different ϵ and B-R at $\Theta_{75} = 7.5°$.

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80% blade span. From $r = 0.80$ to $r = 1.00$, the ALM simulations differ quite substantially from their B-R reference. This difference is attributed to the helical vortex wake of this particular rotor geometry, which sees the tip vortex of the preceding blade hitting the next blade, thus causing a perpendicular BVI at around 93% blade span. This encounter sees the viscous core interacting closely with the blade and induces high velocities as it passes through. The induced velocities are in the plane of the rotor disk and significantly impact the effective angle of attack, which in turn changes the local blade loading. Such a close encounter is not well captured by the method compared to the B-R simulation that has a solid geometry cutting through the previous blade tip vortex, thus creating different flow physics. The force projected by the ALM is porous and, because of its line-like representation, has no way to enforce a nonpenetrating boundary condition as is done in other higher-fidelity methods that rely on source terms such as immersed boundary methods. Toward the tip of the blade, the discrepancies between the various ALM simulations are caused by the differences of the tip vortex structure created by the different force smearing coefficients. Small ϵ values create tighter tip vortices, which in turn create a more realistic induced velocity at the rotor, yielding a more accurate drop off of the thrust blade loading near the tip without having to resort to a tip-loss function that would artificially bring the tip loading down. This behavior is in tune with what is observed in [47], where the ALM simulations with an $\epsilon/c > 0.25$ need correction, whereas values $\epsilon/c \approx 0.25$ produce the most accurate results. The different tip vortex structure has an effect not only on the tip, but also across the whole span of the blade. When considering the inflow angle across the span (not shown), the smaller values of ϵ cause a larger inflow angle. This in turn creates a larger torque loading that is mainly due to induced drag through the rotation of the lift vector by the inflow angle as per BET. The torque loading in Fig. 7 therefore shows a higher torque for the smaller kernel sizes up to 80% blade span. Toward the tip, the BVI described previously changes the shape of the loading compared to the reference B-R case and creates higher drag and torque. Because of the discrepancy created by the BVI that is not properly captured in the tip region, it becomes difficult to decipher which Gaussian smearing produces the best results in this region.

3. *Tip Vortex Characteristics*

The helical vortex wake is of prime importance for the accurate prediction of the rotor loads. The meshes used in this section are quite coarse, which is not ideal when comparing tip vortices as they tend to diffuse quickly. Nonetheless, the vortex positions can be identified for up to a rotor revolution, whereas the tip vortices' radius and circulation are identified for the early wake ages up to approximately 150°, where the tip vortex leaves the ALM refinement zone. Because all ALM simulations are on the same mesh, they can at least be compared to each other on a qualitative basis. The results for the B-R simulation are added for reference purposes only for the vortex locations as their properties would not be comparable to the ALM because of the different meshes used. More specifically, the mesh in the close vicinity of the rotor disk for the B-R simulations is significantly coarser than the ALM, resulting in increased diffusivity. A quantitative analysis of the tip vortex structure and characteristics on a refined mesh is presented in the following section.

Figure 8 presents the evolution of the tip vortex positions over one rotor revolution at a trimmed thrust state of $C_T/\sigma = 0.0723$. The slopes at the top of Fig. 8 represent the radial contraction of the vortices, whereas the ones at the bottom of the graph are the vertical advection below the rotor. In addition, the results of the Kocurek and Tangler empirical model [52] are displayed for reference. All the ALM simulations agree very well and are consistent with both the B-R and empirical model especially for the radial contraction. The axial displacement below the rotor shows a greater variability where the tip vortices from smaller kernels tend to be advected slower. They also better follow the B-R results. Recall that these results are trimmed and thus do not have the same collective settings, which vary considerably. Should the same collective of $\Theta_{75} = 7.5°$ be used, the kernel of $\epsilon/c = 0.25$ would perform the best as its thrust is better predicted across the whole collective sweep without the need of a trim.

The viscous core radius and the circulation of the different tip vortices are presented in Figs. 9a and 9b for wake ages before 150°. The ALM solutions show a general tendency to have a larger vortex core and circulation as ϵ is increased as a consequence of the higher blade loading near the tip for the different ϵ. The only outlier is the ALM with $\epsilon/c = 1.0$, which shows an excessively large vortex core radius and circulation in the first 40.0° of wake age, which then settles to a steadier and more realistic value. This difference is caused by the high blade loading as well as the truncation–normalization of the Gaussian kernel near the tip, which further loads the region at the source term level. This results in a highly elliptical tip vortex in the early wake ages that settles back to a circular profile as the wake age grows. This elliptically shaped tip vortex is not properly captured by the present vortex extraction methodology and empirical vortex

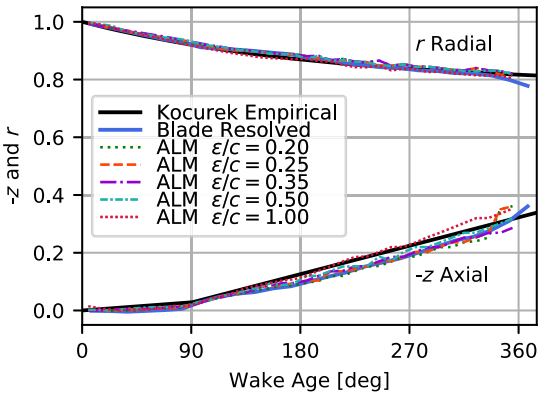
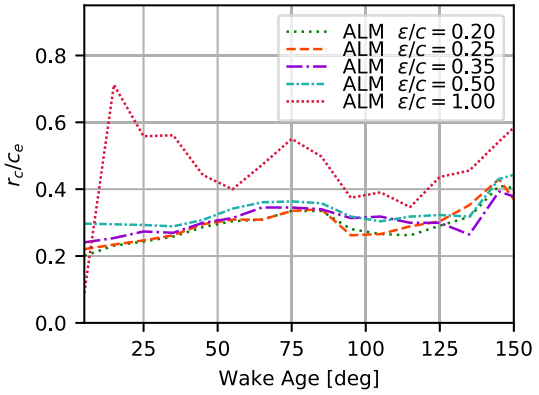
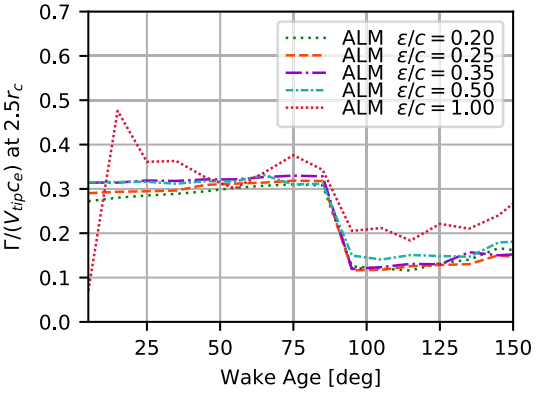


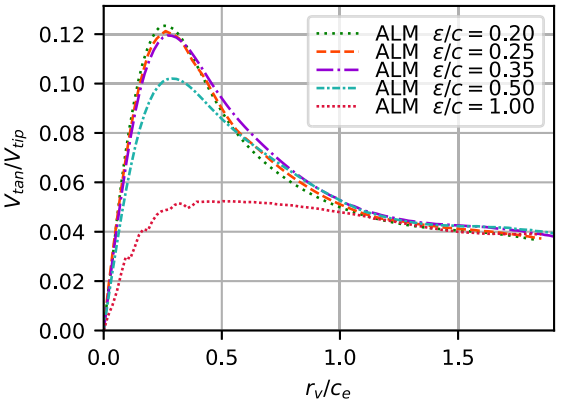
Fig. 8 Tip vortex positions for the ALM at $C_T/\sigma = 0.0723$ and B-R at $\Theta_{75} = 7.5°$.



a) Viscous core radius



b) Vortex circulation



c) Average tangential velocity profile at 35.0° wake age

Fig. 9 Vortex core characteristics in the early wake ages.

model, which causes the high values of the core radius and circulation. The preceding observations are further reflected in Fig. 9c, which shows the average tangential velocity profile of the extracted vortex cores at 35.0° wake age. The position of the maximum tangential velocity of the viscous cores grows as ϵ gets larger. The peak velocity value is highest for $\epsilon/c = 0.20$ and decreases as ϵ is increased. The circulation is computed further along the core, which is where the average velocity gets higher with higher Gaussian kernel sizes, thus explaining the values seen in Fig. 9b. Finally, the shape of the tangential velocity profile for $\epsilon/c = 1.0$ does not fit with the rest of the results or with the empirical model because it is still recovering from its highly elliptical shape.

B. Detailed Comparison

Section V.A shows that lower values of the Gaussian smearing coefficient near the value of $\epsilon/c \approx 0.25$ yield more accurate results as suggested in [45,46]. More specifically, $\epsilon/c = 0.25$ provides better thrust predictive capabilities at a given collective without prior knowledge of the solution, thus preventing the necessity to trim simulations excessively. The value $\epsilon/c = 0.35$ provides an excellent prediction of the torque coefficient and FoM for trimmed solutions. It should be noted that the better agreement of these two integrated coefficients is not a direct consequence of a better blade torque loading from this Gaussian smearing value, but rather from a fortuitous averaging of the BVI disturbed torque loading near the tip. Therefore, this section focuses on a more detailed and quantitative study of the ALM with a selected kernel size of $\epsilon/c = 0.25$ since we value the better thrust predictive capabilities of the method for future applications as well as being consistent with previously published results in the literature. The ALM simulations are run on a similar mesh with respect to their B-R counterparts. The integrated performances plots are generated with their respective medium meshes, whereas the blade loading and tip vortex characteristics are run on the tip vortex refined mesh at a collective trimmed to yield a $C_T/\sigma = 0.09$.

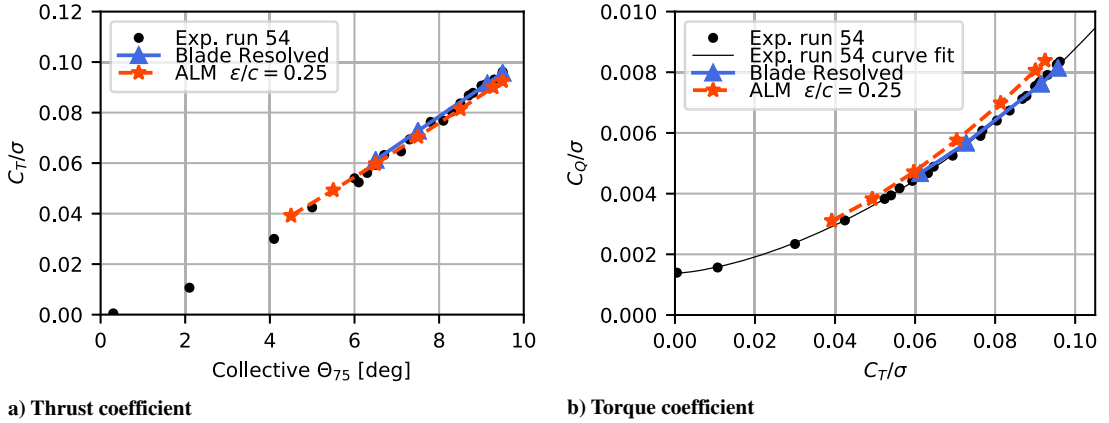
1. *Integrated Performance Coefficients*

The integrated performance plots of the B-R and ALM are presented in Fig. 10. The B-R results show excellent agreement with the experimental data especially at higher collective. For the lowest collective setting, the thrust is slightly overpredicted as well as the torque. This results in a slightly lower FoM. The ALM on the medium mesh performs similarly to the coarse mesh presented in the preceding section. The thrust curve follows a linear trend that overestimates its reference value at lower collective, whereas a slight underprediction is present at the higher collective angles. The torque curve is overestimated across the whole range. When comparing the ALM values to the experimental curve fit of torque coefficient at a given thrust coefficient, the relative error is fairly constant at a value of about 7%. Again, this in turn reflects on an underprediction of the FoM with a constant relative error of about −6.5% compared to the experimental curve fit. For reference, the B-R solutions showed an FoM relative error of −2.74 to 0.9% from the lowest to the highest collective, respectively.

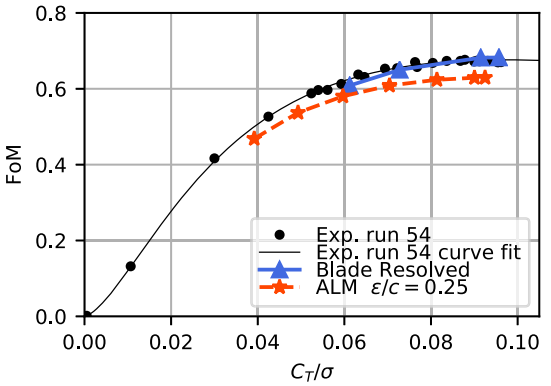
2. *Blade Loading*

The simulations are run at a trimmed state of $C_T/\sigma = 0.09$, which corresponds to a collective angle of $\Theta_{75}^{\text{ALM}} = 9.27°$ and $\Theta_{75}^{\text{BR}} = 9.14°$ for the ALM and B-R simulations, respectively. The results for the blade loading are presented in Fig. 11. In trimmed conditions, the excellent agreement between the two methods for the thrust loading is again shown for the first 80% of the blade span. Then the BVI at around 92% blade span occurs. The ALM overpredicts both the up- and downwash of this direct hit compared to the B-R case, thus causing the discrepancy. It is fortuitous to see that the loss and increase in thrust on both sides of the BVI cancel each other out

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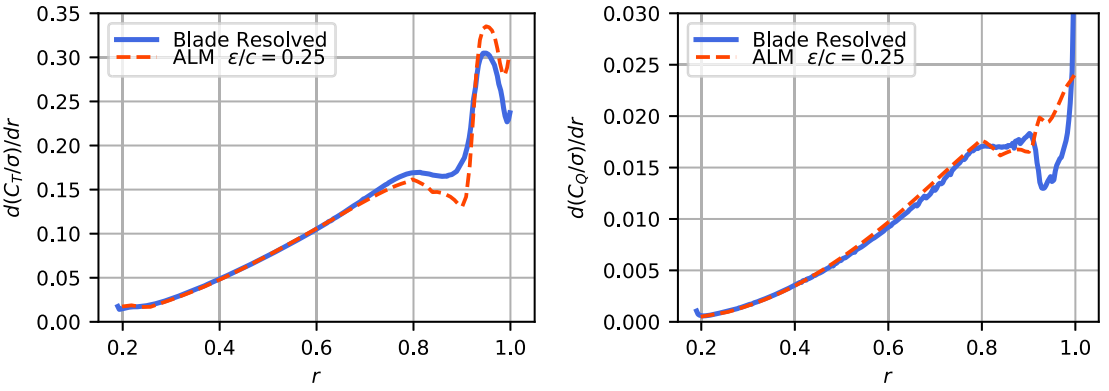


a) Thrust coefficient **b) Torque coefficient**



c) Figure of merit

Fig. 10 Integrated rotor performance coefficients for the ALM and B-R on the medium mesh.



a) Thrust loading **b) Torque loading**

Fig. 11 Sectional blade loading at $C_T/\sigma = 0.09$ for the blade resolved and ALM on the T.V. mesh.

when integrated to recover $C_T/\sigma = 0.09$. The torque loading follows a similar trend, but displays a slight overprediction of torque from span sections $r = 0.5$ to 0.8. The torque loading is again disturbed by the BVI, which causes a rise in sectional induced drag both sides of the vortex impact point. This results in the higher integrated torque coefficient seen in the performance plots.

3. *Tip Vortex Characteristics*

The use of the tip vortex mesh for both the B-R and ALM simulations allows for a more accurate resolutions and quantification of the tip vortex structure and characteristics. Figure 12 shows the positions of the tip vortices over the first rotor revolution. The radial contractions of the two methods are superposed on top of each other as well as the empirical model. The axial advection of the two models is also similar and in good agreement with the prescribed wake model. The slopes of the two simulations are usually lower than the empirical reference, but they both retain the characteristic dual slope before and after the first blade passage with the B-R tip vortex being slightly more advected from 125° to 325° of wake age. Figure 13 presents a qualitative assessment of the wake of the two methods at an isolevel of Q-criterion. The two wakes are very similar in their structure of tip and root vortices, whereas the B-R seems to remain cleaner as a consequence of the smaller time step this simulation uses. A small difference is observed at the first blade passage where the BVI splits the tip vortex of the preceding blade in two. The split vortices have their secondary branch differ as the B-R one is more concentrated and is advected toward the center of the rotor faster than the ALM.

The vortex core properties are presented in Fig. 14 with the vortex core radius on the left and the circulation on the right. The ALM produces a very similar vortex core size to the B-R until the first blade passage, where both methods start with a nondimensional vortex core

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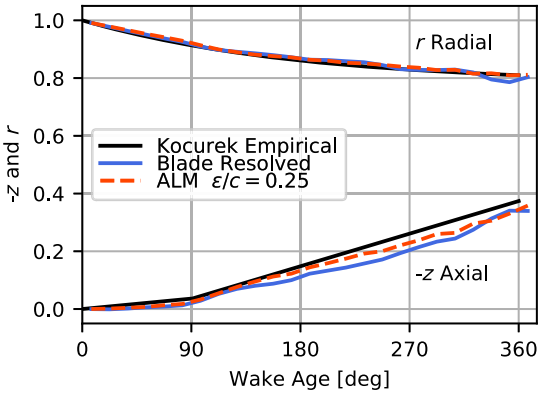


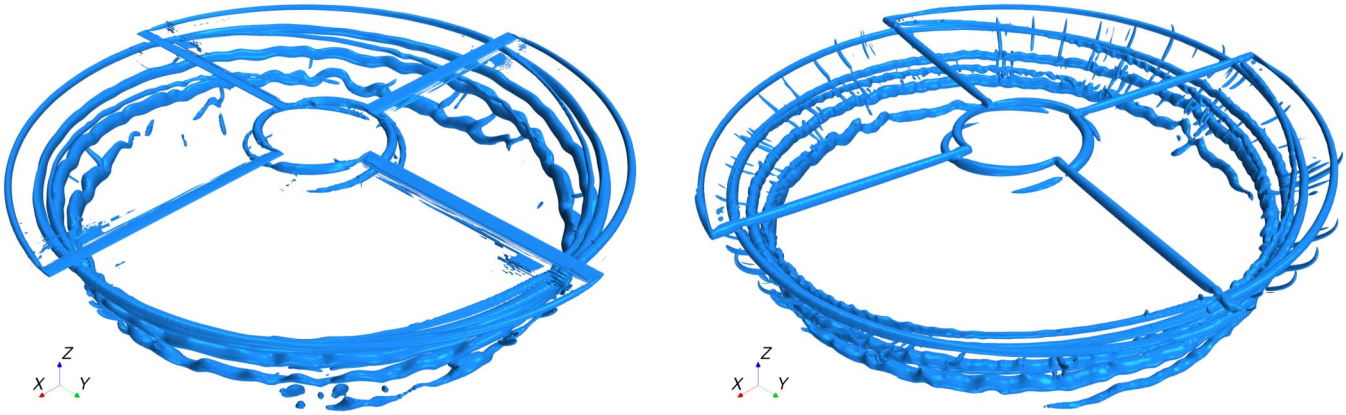
Fig. 12 Tip vortex positions at $C_T/\sigma = 0.09$ for the ALM and B-R on the T.V. mesh.

size of 16% and steadily grow to 26%. After this event, the core radius drops down and starts growing again rapidly due to the second-order scheme employed and its diffusive nature. The circulation follows a similar pattern as the two methods start around a normalized Γ of 0.3 and grow slightly until the first blade passage before dropping significantly and almost halving. Then the vorticity starts to grow again, but remains more constant than the vortex core radius. The difference between the two slopes before the first blade passage is attributed to the higher peak blade loading the ALM has over the B-R solution. Figure 15 shows the average tangential velocity plots of the two methods for wake ages under the first two blade passages. They complement the preceding figure by showing that the core velocity profile is higher for the B-R at the center of the core radius even though the circulation of the ALM is stronger for the first 90° of wake age. That peak velocity decays faster in the case of the B-R solution in that interval. Then after the first blade passage, the BVI considerably reduces the circulation of the vortices with the ALM being less affected. After the BVI, the vortex core of the B-R grows at a faster rate than the ALM.

The growing discrepancies between the two methods after the first blade passage is explained by the different response the two methods have toward the BVI. Figure 16 presents the vorticity magnitude in slices from the extraction planes at wake ages 2.5° prior and 5°, 10°, and 15° after the first blade passage. At 87.5°, both methods are similar, with the ALM showing a slightly higher vorticity peak. Then, at 5° past the blade passage, the two methods show a secondary vortical structure that is also present in Fig. 13. In both cases, the vorticity and size of the primary vortical structure decreases from before the BVI. The secondary vortex between the two methods differs as the ALM creates a weaker and larger vortex compared to the B-R. Then the two vortices evolve differently as the secondary vortex of the B-R simulation tends to be advected toward the center of the rotor faster than its ALM counterpart. As already mentioned in Sec. V.A.2, the ALM projects the force in a porous manner and has no means to enforce a nonpenetrating boundary conditions, thus leading to a different response when a BVI occurs.

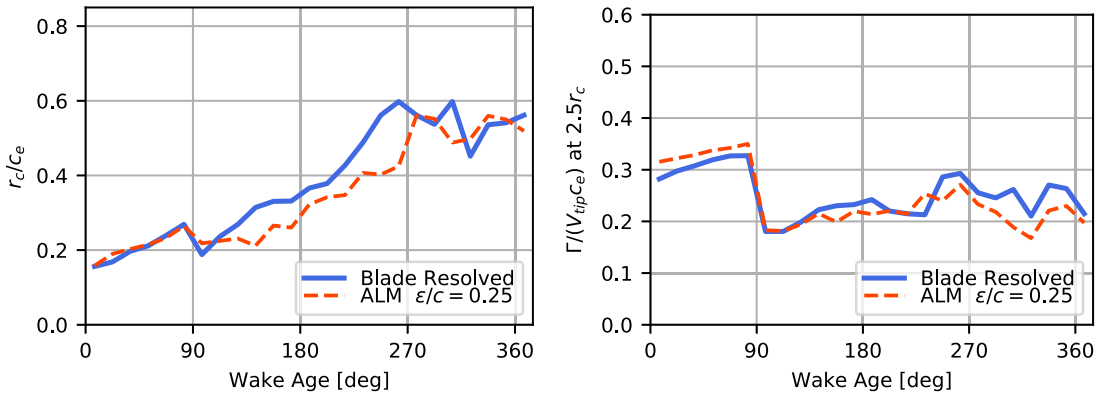
4. *Computational Time*

The parallel scaling efficiency of the two methods is presented in Fig. 17. This strong scaling study (same mesh, varying amount of cores) is run on the Béluga cluster of Compute Canada, which features 40 core nodes composed of two Intel Xeon Gold 6148



a) B-R **b) ALM**

Fig. 13 Q-criterion at isolevel $Q = 25{,}000$ for $C_T/\sigma = 0.09$.



a) Viscous core radius **b) Vortex circulation**

Fig. 14 Vortex core characteristics over one rotor revolution at $C_T/\sigma = 0.09$.

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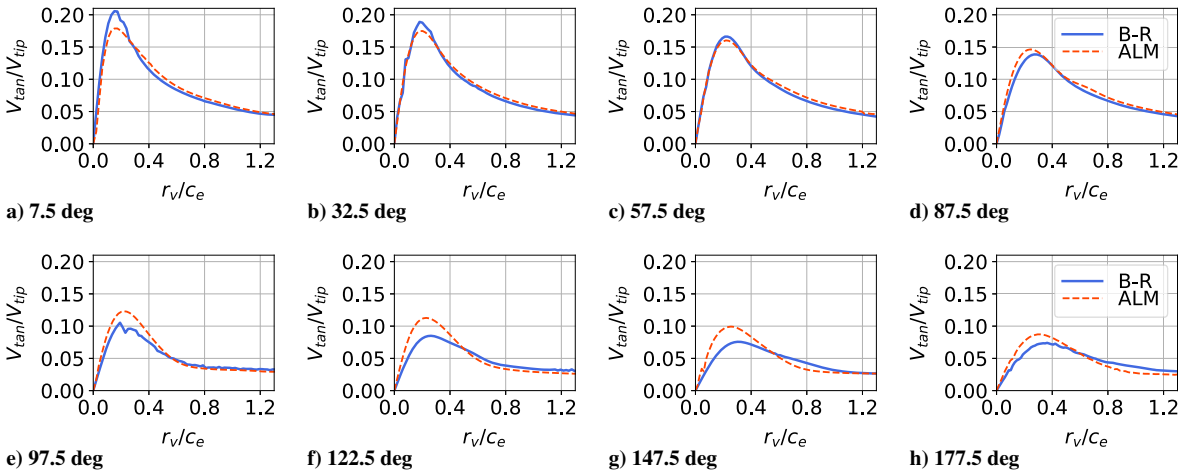


Fig. 15 Vortex core velocity profiles over two blade passages at $C_T/\sigma = 0.09$.

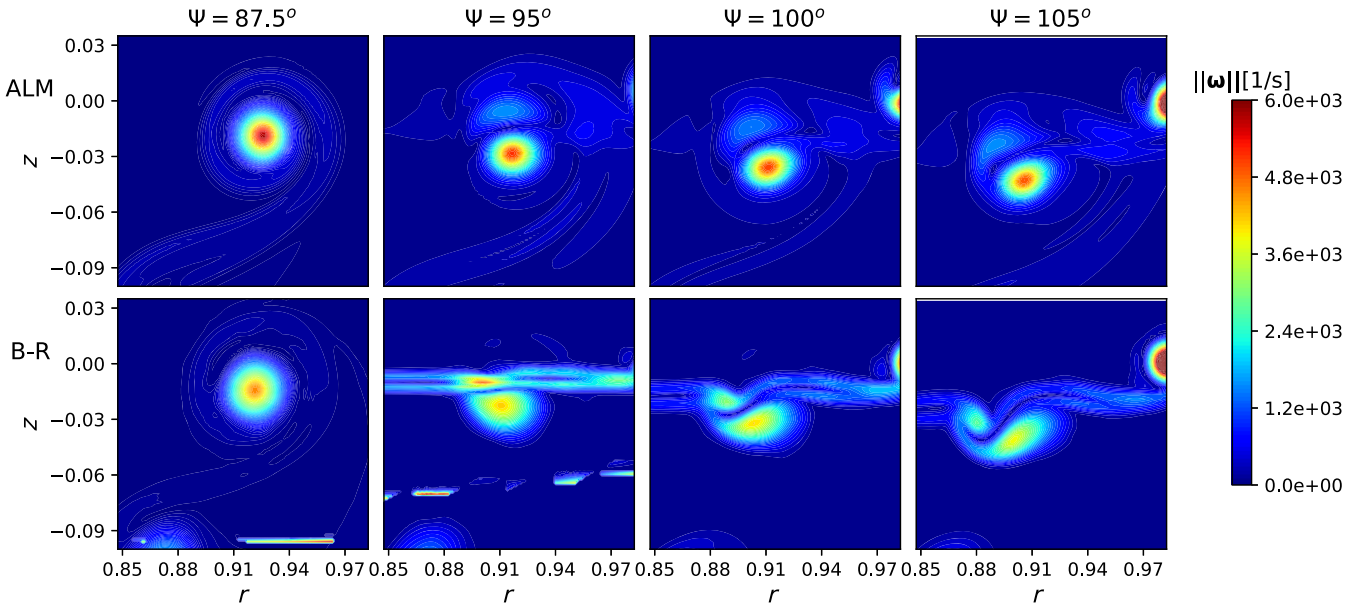


Fig. 16 Vorticity magnitude slices at $C_T/\sigma = 0.09$.

processors each. The compute times of the two methods are measured as the average time over 300 iterations and then normalized on a CPUtime[s]/(iteration × Nb_{cell}) basis. The compute times are compared to their respective one node reference time. The B-R on the coarse mesh and the ALM on the medium mesh are used. The speed-up curves are presented as a function of the number of cores used. Because the two methods have a different amount of mesh cells, an additional metric is included. The amount of mesh cells per core is presented on a secondary and tertiary x axes as they offer a more representative evaluation of performances. The B-R simulations quickly lose their parallel efficiency as the curves drop below 75% at 75,000 cells per core as a consequence of the poor scaling capabilities of the moving overset module. The ALM, on the other hand, is run on a single mesh and sees its speed-up scale much better with a parallel efficiency of 80% up to 10,500 cells per core, which is comparable to StarCCM+ run on a single mesh region.

Table 4 reports the computational times of the two methods over the four meshes used in this study as well as the speed-up the ALM has with respect to the B-R. The total compute times are averaged over the span of the whole simulation and transformed in CPU time per revolution. As seen by the results, the ALM is on average 3–4 times quicker than an equivalent B-R simulation on similar background meshes. These time improvements are largely attributed to

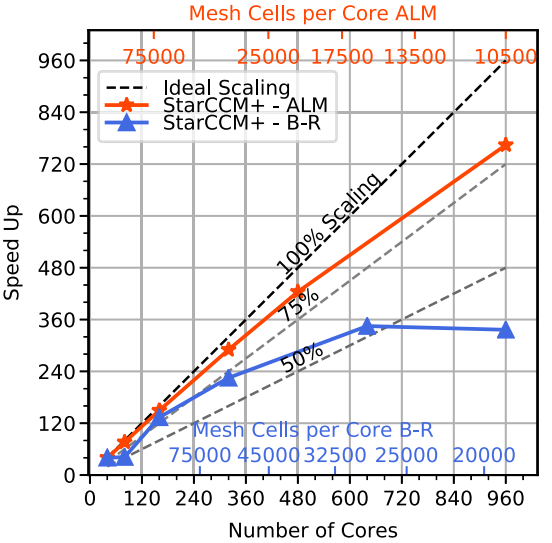


Fig. 17 Parallel scaling efficiency of the ALM and blade resolved codes on Béluga.

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Table 4 Compute time averaged for one revolution and speed-up

	CPU time per revolution [h]			
	Coarse	Medium	Fine	T.V.
B-R	1,219.8	2,321.5	7,362.0	23,813.5
ALM	284.8	741.9	1,771.5	5,192.5
Speed-up	4.3	3.1	4.2	4.6

the advantageous mesh size an ALM simulation has over the B-R solution.

VI. Conclusions

An actuator line model implementation has been developed and adapted for helicopter rotor flow simulations in hover. Key improvements to the method include a modified Gaussian kernel that is truncated and normalized, leading to a more consistent force projection at smaller ϵ smearing values. The hyperbolic tangent span discretization enables finer details to be captured at the tip, while the force under-relaxation and ramp start increase the stability and convergence of the method substantially.

Proper selection of the kernel width ϵ is explored with a parametric sweep around its optimal value of $\epsilon/c = 0.25$. It is shown that the optimal kernel produces the best overall results and manages to naturally recover a proper behavior of the blade loading at the tip, which is often not the case with other ALM implementations that use larger kernel sizes and rely on tip-loss functions. The larger kernels produce larger tip vortex cores and higher circulation, which in turn changes the induced velocity the blade sees, resulting in higher blade loading near the tip and higher thrust coefficients at a given collective. Furthermore, the different inflow creates different induced drag characteristics across the span of the blade where smaller kernels see a higher drag resulting in higher torque.

When trimmed to an identical thrust value on equivalent background meshes, the ALM produces similar results compared to its B-R counterpart. The only outstanding issue is the overprediction of the induced velocity a perpendicular BVI causes as the porosity of the ALM force projection does not allow the same interaction between the blade and the vortex when compared to the B-R case. Fortuitously, the thrust under- and overestimation near the blade tip region cancel out when looking at integrated thrust results, but it causes an increase in drag, resulting in torque coefficient and FoM curves with an offset. The tip vortex trajectories and core characteristics are in good agreement.

Finally, the computational efficiency of the ALM is showcased. The excellent scalability and 3–4× speed-up across all meshes show its potential in simulations where main flow features and rotor response are of interest and computational efficiency is needed.

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References

[1] Hoencamp, A., van Holten, T., and Prasad, J. V. R., "Relevant Aspects of Helicopter-Ship Operations," *Proceedings of the 34th European Rotorcraft Forum ERF34*, Vol. 1, Royal Aeronautical Soc., Liverpool, England, U.K., 2008, pp. 578–588.

[2] Hodge, S. J., Zan, S. J., Roper, D. M., Padfield, G. D., and Owen, I., "Time-Accurate Ship Airwake and Unsteady Aerodynamic Loads Modeling for Maritime Helicopter Simulation," *Journal of the American Helicopter Society*, Vol. 54, No. 2, 2009, Paper 022005. https://doi.org/10.4050/JAHS.54.022005

[3] Roper, D. M., Owen, I., Padfield, G. D., and Hodge, S. J., "Integrating CFD and Piloted Simulation to Quantify Ship-Helicopter Operating Limits," *Aeronautical Journal*, Vol. 110, No. 1109, 2006, pp. 419–428. https://doi.org/10.1017/S0001924000001329

[4] Crozon, C., Steijl, R., and Barakos, G. N., "Numerical Study of Helicopter Rotors in a Ship Airwake," *Journal of Aircraft*, Vol. 51, No. 6, 2014, pp. 1813–1832. https://doi.org/10.2514/1.C032535

[5] Oruc, I., Horn, J. F., Polsky, S., Shipman, J., and Erwin, J., "Coupled Flight Dynamics and CFD Simulations of the Helicopter/Ship Dynamic Interface," *American Helicopter Society 71st Annual Forum*, AHS International, Virginia Beach, VA, 2015.

[6] Johnson, W., *Helicopter Theory*, Dover, New York, 1994.

[7] Zagaglia, D., Giuni, M., and Green, R. B., "Rotor-Obstacle Aerodynamic Interaction in Hovering Flight: An Experimental Survey," *American Helicopter Society 72nd Annual Forum*, AHS International, West Palm Beach, FL, 2016.

[8] Quackenbush, R., Chua, K., Wachspress, D. A., and Boschitsch, H., "Computation of Rotor Aerodynamic Loads in Forward Flight Using a Full-Span Free Wake Analysis," NASA CR 77611, Oct. 1990.

[9] Ferlisi, C., "Rotor Wake Modelling Using the Vortex-Lattice Method," Master's Thesis, Ecole Polytechnique Montreal, Montreal, April 2018.

[10] Katz, J., and Plotkin, A., *Low-Speed Aerodynamics*, 2nd ed., Cambridge Univ. Press, New York, 2001.

[11] Hariharan, N. S., Egolf, T. A., and Sankar, L. N., "Simulation of Rotor in Hover: Current State, Challenges and Standardized Evaluation," *52nd Aerospace Sciences Meeting*, AIAA Paper 2014-0041, 2014. https://doi.org/10.2514/6.2014-0041

[12] Steijl, R., and Barakos, G., "CFD Analysis of Complete Helicopter Configurations–Lessons Learnt from the GOAHEAD Project," *Aerospace Science and Technology*, Vol. 19, No. 1, 2012, pp. 58–71. https://doi.org/10.1016/j.ast.2011.01.007

[13] Jimenez Garcia, A., and Barakos, G. N., "Accurate Predictions of Rotor Hover Performance at Low and High Disc Loadings," *Journal of Aircraft*, Vol. 55, No. 1, 2018, pp. 89–110. https://doi.org/10.2514/1.C034144

[14] Sørensen, J. N., and Shen, W. Z., "Numerical Modeling of Wind Turbine Wakes," *Journal of Fluids Engineering*, Vol. 124, No. 2, 2002, pp. 393–399. https://doi.org/10.1115/1.1471361

[15] Rajagopalan, R. G., and Mathur, S. R., "Three Dimensional Analysis of a Rotor in Forward Flight," *Journal of the American Helicopter Society*, Vol. 38, No. 3, 1993, pp. 14–25. https://doi.org/10.4050/JAHS.38.14

[16] Khier, W., "Time-Accurate Versus Actuator Disk Simulations of Complete Helicopters," *High Performance Computing in Science and Engineering' 05*, edited by W. E. Nagel, M. Resch, and W. Jäger, Springer–Verlag, Berlin, 2006, pp. 209–220. https://doi.org/10.1007/3-540-29064-8_16

[17] O'Brien, D. M., Jr., and Smith, M. J., "Analysis of Rotor-Fuselage Interactions Using Various Rotor Models," *43rd AIAA Aerospace Sciences Meeting*, AIAA Paper 2005-0468, 2005. https://doi.org/10.2514/6.2005-468

[18] Forsythe, J. R., Lynch, E., Polsky, S., and Spalart, P., "Coupled Flight Simulator and CFD Calculations of Ship Airwake Using Kestrel," *53rd AIAA Aerospace Sciences Meeting*, AIAA Paper 2015-0556, 2015. https://doi.org/10.2514/6.2015-0556

[19] Schmitz, S., and Jha, P. K., "Modeling the Wakes of Wind Turbines and Rotorcraft Using the Actuator-Line Method in an OpenFOAM—LES Solver," *American Helicopter Society 69th Annual Forum*, AHS International, Phoenix, Arizona, 2013.

[20] Stanly, R., Delorme, Y. T., and Frankel, S. H., "Computational Assessment of Actuator Line Model for Large Eddy Simulation of Rotor Noise," *AIAA Scitech 2020 Forum*, AIAA Paper 2020-0035, 2020. https://doi.org/10.2514/6.2020-0035

[21] Alpman, E., Long, L. N., Bridges, D. O., and Horn, J. F., "Fully-Coupled Simulations of the Rotorcraft/Ship Dynamic Interface," *American Helicopter Society International 63rd Annual Forum*, AHS International, Virginia Beach, VA, 2007, pp. 1367–1382.

[22] Boyd, D. D., Jr., "Rotor/Fuselage Unsteady Interactional Aerodynamics: A New Computational Model," Ph.D. Thesis, Virginia Tech, Blacksburg, VA, 1999.

[23] Chirico, G., Szubert, D., Vigevano, L., and Barakos, G. N., "Numerical Modelling of the Aerodynamic Interference Between Helicopter and Ground Obstacles," *CEAS Aeronautical Journal*, Vol. 8, No. 4, 2017, pp. 589–611. https://doi.org/10.1007/s13272-017-0259-y

[24] O'Brien, D. M., Jr., and Smith, M. J., "Understanding the Physical Implications of Approximate Rotor Methods Using an Unstructured CFD Method," *Proceedings of the 31st European Rotorcraft Forum ERF31*, Vol. 1, Associazione Italiana de Aerotecnica e Astronautica

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(AIDAA), Florence Italy, 2005, pp. 798–809.
https://doi.org/20.500.11881/1162

[25] Kim, Y.-H., and Park, S.-O., "Navier-Stokes Simulation of Unsteady Rotor-Airframe Interaction with Momentum Source Method," *International Journal of Aeronautical and Space Sciences*, Vol. 10, No. 2, 2009, pp. 125–133.
https://doi.org/10.5139/IJASS.2009.10.2.125

[26] Kim, Y. H., and Park, S. O., "Unsteady Momentum Source Method for Efficient Simulation of Rotor Aerodynamics," *Journal of Aircraft*, Vol. 50, No. 1, 2013, pp. 324–327.
https://doi.org/10.2514/1.C031934

[27] Wood, S. L., Coder, J. G., and Hariharan, N. S., "Fundamental Investigation of Grid Resolution on Wake Fidelity Submitted to the APATC Special Session: Simulation of Rotor in Hover," *2018 AIAA Aerospace Sciences Meeting*, AIAA Paper 2018-0307, 2018.
https://doi.org/10.2514/6.2018-0307

[28] Kim, T., Oh, S., and Yee, K., "Novel Actuator Surface Method for Helicopter Rotor Analysis," *Journal of Aircraft*, Vol. 53, No. 6, 2016, pp. 1947–1952.
https://doi.org/10.2514/1.C033666

[29] Linton, D., Barakos, G., Widjaja, R., and Thornbern, B., "A New Actuator Surface Model with Improved Wake Model for CFD Simulations of Rotorcraft," *American Helicopter Society 73rd Annual Forum*, AHS International, Fort Worth, TX, 2017.

[30] Linton, D., Barakos, G., Widjaja, R., and Thornber, B., "Coupling of an Unsteady Aerodynamics Model with a Computational Fluid Dynamics Solver," *AIAA Journal*, Vol. 56, No. 8, 2018, pp. 3153–3166.
https://doi.org/10.2514/1.J056784

[31] Balch, D. T., and Lombardi, J., "Experimental Study of Main Rotor Tip Geometry and Tail Rotor Interactions in Hover Vol 2—Run Log and Tabulated Data," NASA CR 177336, 1985.

[32] Balch, D. T., and Lombardi, J., "Experimental Study of Main Rotor Tip Geometry and Tail Rotor Interactions in Hover Vol 1—Text and Figures," NASA CR 177336, 1985.

[33] Bousman, W. G., "Aerodynamic Characteristics of SC1095 and SC1094 R8 Airfoils," NASA TP 212265, Defense Technical Information Center, Fort Belvoir, VA, Dec. 2003.
https://doi.org/10.21236/ADA480203

[34] Jain, R., "Hover Predictions on the S-76 Rotor with Tip Shape Variation Using Helios," *Journal of Aircraft*, Vol. 55, No. 1, 2018, pp. 66–77.
https://doi.org/10.2514/1.C034075

[35] Bourgault-Côté, S., Ghasemi, S., Mosahebi, A., and Laurendeau, E., "Extension of a Two-Dimensional Navier–Stokes Solver for Infinite Swept Flow," *AIAA Journal*, Vol. 55, No. 2, 2017, pp. 662–667.
https://doi.org/10.2514/1.J055139

[36] Parenteau, M., Sermeus, K., and Laurendeau, E., "VLM Coupled with 2.5D RANS Sectional Data for High-Lift Design," *2018 AIAA Aerospace Sciences Meeting*, AIAA Paper 2018-1049, 2018.
https://doi.org/10.2514/6.2018-1049

[37] "STAR-CCM+ Documentation," User Manual Version 12.06, Siemens PLM Software, 2016, https://community.sw.siemens.com/s/question/0D54O000075Nz8xSAC/how-do-i-cite-simcenter-starccm.

[38] Weiss, J. M., Maruszewski, J. P., and Smith, W. A., "Implicit Solution of Preconditioned Navier-Stokes Equations Using Algebraic Multigrid," *AIAA Journal*, Vol. 37, No. 1, 1999, pp. 29–36.
https://doi.org/10.2514/2.689

[39] Menter, F. R., "Two-Equation Eddy-Viscosity Turbulence Models for Engineering Applications," *AIAA Journal*, Vol. 32, No. 8, 1994, pp. 269–289.
https://doi.org/10.2514/3.12149

[40] Mohd, N. A. R. N., and Barakos, G. N., "Computational Aerodynamics of Hovering Helicopter Rotors," *Jurnal Mekanikal*, No. 34, June 2012, pp. 16–46.

[41] Vieira, B. A., Kinzel, M. P., and Maughmer, M. D., "CFD Hover Predictions Including Boundary-Layer Transition," *55th AIAA Aerospace Sciences Meeting*, AIAA Paper 2017-1665, 2017.
https://doi.org/10.2514/6.2017-1665

[42] Merabet, R., and Laurendeau, E., "Parametric Study on the Velocity Sampling Techniques for the Actuator Line Method in 2D," *AIAA SciTech 2019 Forum*, AIAA Paper 2019-1797, 2019.
https://doi.org/10.2514/6.2019-1797

[43] Churchfield, M. J., Schreck, S. J., Martinez, L. A., Meneveau, C., and Spalart, P. R., "An Advanced Actuator Line Method for Wind Energy Applications and Beyond," *35th Wind Energy Symposium*, AIAA Paper 2017-1998, 2017.
https://doi.org/10.2514/6.2017-1998

[44] Caprace, D.-G., Chatelain, P., and Winckelmans, G., "Lifting Line with Various Mollifications: Theory and Application to an Elliptical Wing," *AIAA Journal*, Vol. 57, No. 1, 2019, pp. 17–28.
https://doi.org/10.2514/1.J057487

[45] Martinez-Tossas, L. A., Churchfield, M. J., and Meneveau, C., "Optimal Smoothing Length Scale for Actuator Line Models of Wind Turbine Blades Based on Gaussian Body Force Distribution," *Wind Energy*, Vol. 20, No. 6, 2017, pp. 1083–1096.
https://doi.org/10.1002/we.2081

[46] Shives, M., and Crawford, C., "Mesh and Load Distribution Requirements for Actuator Line CFD Simulations," *Wind Energy*, Vol. 16, No. 8, 2012, pp. 1183–1196.
https://doi.org/10.1002/we.1546

[47] Martinez-Tossas, L. A., and Meneveau, C., "Filtered Lifting Line Theory and Application to the Actuator Line Model," *Journal of Fluid Mechanics*, Vol. 863, March 2019, pp. 269–292.
https://doi.org/10.1017/jfm.2018.994

[48] Nathan, J., Masson, C., and Dufresne, L., "Optimal Distribution Width for ALM in LES for (NEW) MEXICO Experiment," *Journal of Physics: Conference Series*, Vol. 1037, June 2018, Paper 072026.
https://doi.org/10.1088/1742-6596/1037/7/072026

[49] Jha, P. K., Churchfield, M. J., Moriarty, P. J., and Schmitz, S., "Guidelines for Volume Force Distributions Within Actuator Line Modeling of Wind Turbines on Large-Eddy Simulation-Type Grids," *Journal of Solar Energy Engineering*, Vol. 136, No. 3, 2014, Paper 031003.
https://doi.org/10.1115/1.4026252

[50] Jain, R., "Sensitivity Study of High-Fidelity Hover Predictions on the Sikorsky S-76 Rotor," *Journal of Aircraft*, Vol. 55, No. 1, 2018, pp. 78–88.
https://doi.org/10.2514/1.C034076

[51] Goerttler, A., Braukmann, J. N., Schwermer, T., Gardner, A. D., and Raffel, M., "Tip-Vortex Investigation on a Rotating and Pitching Rotor Blade," *Journal of Aircraft*, Vol. 55, No. 5, 2018, pp. 1792–1804.
https://doi.org/10.2514/1.C034693

[52] Kocurek, J. D., and Tangler, J. L., "A Prescribed Wake Lifting Surface Hover Performance Analysis," *Journal of the American Helicopter Society*, Vol. 22, No. 1, 1977, pp. 24–35.
https://doi.org/10.4050/JAHS.22.24

[53] Vatistas, G. H., Kozel, V., and Mih, W. C., "A Simpler Model for Concentrated Vortices," *Experiments in Fluids*, Vol. 11, No. 1, 1991, pp. 73–76.
https://doi.org/10.1007/BF00198434

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